

Statistics for Health Data Science - Assessment 2

Student ID: 14210037

1) Summarise the data (6 marks)

This study analysed 997 patients randomly collected from clinical settings in Greater Manchester to identify risk factors for the early prevention of dementia. The sample comprised 945 individuals without dementia (94.8%) and 52 with confirmed dementia (5.2%), providing sufficient sample size for statistical inference.

Table 1. Baseline characteristics stratified by dementia status

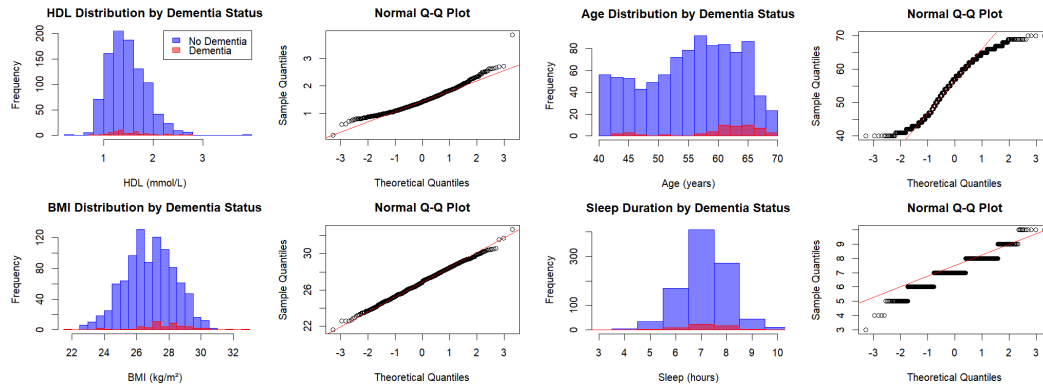
	Overall	Dementia Status	
		No Dementia	Dementia
n	997	945	52
HDL (mmol/L)	1.47 (0.38)	1.47 (0.38)	1.52 (0.40)
BMI (kg/m ²)	26.87 (1.57)	26.83 (1.54)	27.70 (1.91)
Age (mean (SD))	56.08 (7.92)	55.79 (7.88)	61.42 (6.78)
Sleep (hours)			
3	1 (0.1%)	1 (0.1%)	0 (0.0%)
4	4 (0.4%)	4 (0.4%)	0 (0.0%)
5	36 (3.6%)	34 (3.6%)	2 (3.8%)
6	178 (17.9%)	170 (18.0%)	8 (15.4%)
7	432 (43.3%)	409 (43.3%)	23 (44.2%)
8	289 (29.0%)	272 (28.8%)	17 (32.7%)
9	47 (4.7%)	45 (4.8%)	2 (3.8%)
10	10 (1.0%)	10 (1.1%)	0 (0.0%)
Irritability, n (%)			
No	709 (71.1%)	669 (70.8%)	40 (76.9%)
Yes	288 (28.9%)	276 (29.2%)	12 (23.1%)
Physical activity, n (%)			
≤15 min	34 (3.4%)	33 (3.5%)	1 (1.9%)
>15 min	963 (96.6%)	912 (96.5%)	51 (98.1%)
Smoking, n (%)			
Never	584 (58.6%)	558 (59.0%)	26 (50.0%)
Ex-smoker	343 (34.4%)	325 (34.4%)	18 (34.6%)
Current	70 (7.0%)	62 (6.6%)	8 (15.4%)
Dementia diagnosis, n (%)			
No	931 (93.4%)	927 (98.1%)	4 (7.7%)
Yes	66 (6.6%)	18 (1.9%)	48 (92.3%)

Note: Continuous variable: mean ± SD; categorize variable : n (%)

Table 1 presents baseline characteristics stratified by dementia status. High-density lipoprotein (HDL) cholesterol level averaged 1.47 mmol/L (std = 0.38), within the healthy range (>1.0 mmol/L for men; >1.2 mmol/L for women) according to NHS guidelines. The population's mean Body Mass Index (BMI) was 26.9 kg/m² (standard deviation, std = 1.6), falling within the overweight range (25.0-29.9 kg/m²). The average age was middle adulthood, 56.1 years old (std = 7.9) as defined by WHO criteria. Regarding lifestyle factors, self-reported sleep duration ranged from 3 to 10 hours daily, with 7 hours being most common (43.3%). The majority of participants (96.6%, n=963) reported engaging in daily physical activity exceeding 15 minutes. Irritability was present in 28.9% (n=288) of the sample. For smoking status, 58.6% (n=584) were never-smokers, 34.4% (n=343) were ex-smokers, and 7.0% (n=70) were current smokers.

a. Continuous variables

Showing the distribution by histograms grouped by true dementia status, supported by QQ plots to illustrate the normality.



b. Categories variables

Demonstrate the distribution by boxplots grouped by true dementia status

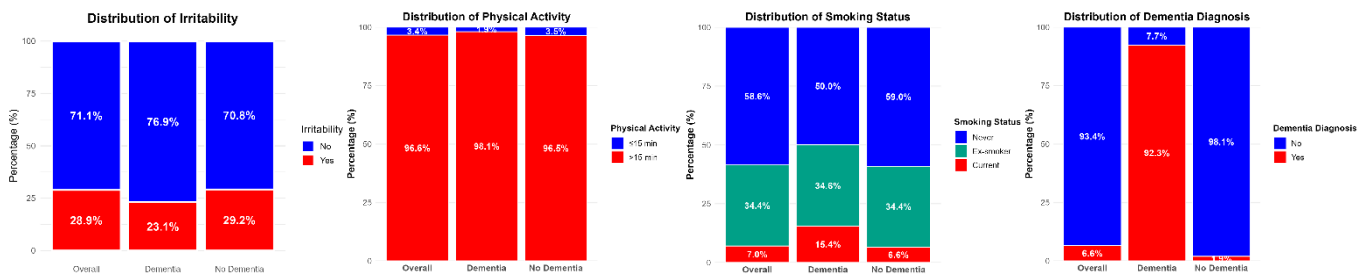


Figure 1. The observation distribution under different variables.

Figure 1 illustrates the distribution of potential risk factors. Among continuous variables, only BMI shows normal distribution. HDL was right-skewed and age were higher around 55 to 65 years. Sleeping hours, captured in integer values, were considered a pseudo-continuous variable. For categorical variables, more than 70% reported no irritability in both groups, and over 95% indicated daily exercise exceeding 15 minutes. Current smoking was twice as common in dementia groups compared with non-dementia groups, with nearly equal proportions of ex-smokers. Regarding diagnosis, 7.7% in dementia groups and 1.9% in non-dementia groups were misdiagnosed.

In conclusion, age, BMI, and smoking status may influence dementia prevention, warranting further study through multivariable modeling (Question 3). Although HDL, sleep, irritability, and physical activity showed no significant group differences, these factors may still contribute to dementia prevention.

2) Do the average levels of high-density lipoprotein differ across smoking groups?

This study will focus on HDL cholesterol levels across different smoking statuses; thus, one-way ANOVA was conducted. The confidence interval was set at 95%, implying that Type 1 error rate of 5% is accepted. In other words, there remains a 5% probability of

incorrectly rejecting it. Prior to statistical testing, the assumption of independence was fulfilled by the study design, with each patient contributing one observation. Results of the Shapiro-Wilk normality test indicated that the data deviated from normality in non-smokers and ex-smokers (both $p < 0.05$). Therefore, the original one-way ANOVA could not be applied, necessitating the use of the non-parametric Kruskal-Wallis rank sum test to compare median HDL levels across groups. The Kruskal-Wallis test evaluates the median rather than the mean, which more reliably represents the dataset.

```
> by(d$HDL, d$Smoking, shapiro.test)
```

```
d$Smoking: 0 W = 0.96427, p-value = 1.021e-10
```

```
d$Smoking: 1 W = 0.96904, p-value = 1.079e-06
```

```
d$Smoking: 2 W = 0.97175, p-value = 0.1145
```

```
# H0: HDL median levels are equal across all Smoking groups.
```

```
# H1: At least one group differs in median HDL level.
```

```
> kruskal.test(HDL ~ Smoking, data = d)
```

```
Kruskal-Wallis chi-squared = 5.6717, df = 2, p-value = 0.05867
```

The result did not provide enough evidence to reject the null hypothesis ($\chi^2 = 5.67$, $df = 2$, $p\text{-value} = 0.059$), meaning HDL median levels were no different between smoking statuses. However, the p-value approaching the significance threshold suggests a potential Type II error, where a true difference might exist but was not detected due to limited sample size, particularly in the current smoker group ($n=70$). Thus, the pairwise Wilcox test, with Bonferroni adjustment, and calculation of the 95% CIs for each group's median were conducted.

```
> pairwise.wilcox.test(d$HDL, d$Smoking, p.adjust.method = "bonferroni")
```

Pairwise Wilcoxon rank-sum tests indicated no statistically significant differences in HDL cholesterol levels across smoking categories: non-smoker vs. ex-smoker ($p = 0.639$), current smoker vs. non-smoker ($p = 0.081$), and current smoker vs. ex-smoker ($p = 0.354$). Median estimates with 95% confidence intervals further support these findings: non-smokers (median = 1.465, CI = [1.418, 1.500]), ex-smokers (median = 1.401, CI = [1.352, 1.470]), and current smokers (median = 1.3465, CI = [1.2590, 1.4510]). The substantial overlap among confidence intervals underscores the absence of differences between groups. Collectively, these results suggest that HDL cholesterol levels do not differ appreciably across smoking status in this population.

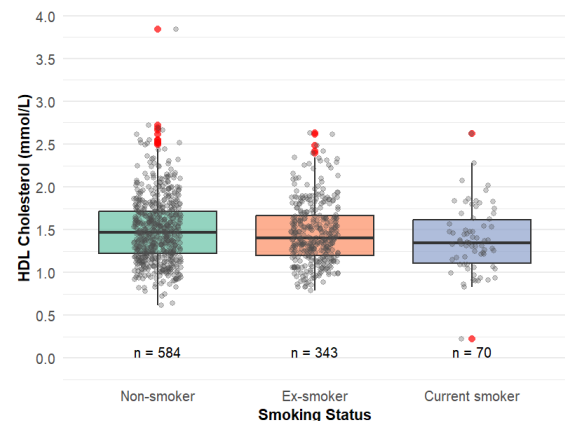
Table 2 shows the HDL levels separated by smoking history. Non-smokers had a median HDL of 1.45 mmol/L (IQR: 1.20-1.70), ex-smokers 1.48 mmol/L (IQR: 1.23-1.73), and current smokers 1.35 mmol/L (IQR: 1.15-1.60). Figure 2 visualises the data. The medians in each group are close. Moreover, the narrow IQR suggests tightly clustered within groups.

In summary, while the HDL cholesterol level median was lower in the smokers (1.35

mmol/L), the difference across smoking statuses did not reach statistical significance (p=0.059).

Table 2. HDL Cholesterol Levels by Smoking Status

Smoking	N	Median (IQR)	Min	Max
Non-smoker	584	1.47 (1.22 - 1.71)	0.62	3.84
Ex-smoker	343	1.40 (1.20 - 1.66)	0.79	2.63
Current smoker	70	1.35 (1.11 - 1.61)	0.23	2.62



3) Explore risk factors of dementia. (For this task, you don't need to state and check the assumptions required in your selected statistical model or test. 30 marks.)

To explore the outcome of the potential parameters for true dementia, the purpose is to find the most fit linear models within the collected variables. Because the outcome is getting dementia disease or not (binomial), the logistic regression, using logit to transform to linear, was conducted. Thus, the assumption of the data needs to be checked, but is omitted for limited pages for this assessment.



3.1. Full Model Analysis

```
> d2 <- d[, !(names(d) %in% c("ID", "Dementia_diag"))]
```

```
> full_m <- glm(Dementia ~ ., data = d2, family = binomial(link = "logit"))
```

Coefficients:	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-21.29075	3.42440	-6.217	5.06e-10 ***
HDL	0.01011	0.37968	0.027	0.97877
Sleep	-0.19511	0.15923	-1.225	0.22046
BMI	0.43383	0.10682	4.061	4.88e-05 ***
Age	0.12518	0.02529	4.950	7.44e-07 ***
IrritabilityYes	-0.21933	0.35160	-0.624	0.53275
SmokingEx-smoker	0.02597	0.32722	0.079	0.93673
SmokingCurrent	1.47658	0.46177	3.198	0.00139 **
Physical_activity>15 min	0.46844	1.06411	0.440	0.65978

Null deviance: 408.4 on 996 degrees of freedom

Residual deviance: 352.9 on 988 degrees of freedom **AIC: 370.9**

The summary for the full model showed that BMI and age are statistically significant (p < 0.001), with current smoking also significant (p < 0.05). While other covariates did not reach

the significance level, this indicates that not all predictors contribute meaningfully to dementia risk prediction. Model fit was shown on the Akaike Information Criterion (AIC), equal to 370.9.

3.2. Model selection strategies

The AIC was used for this model selection because it would not omit possible variables in medium sample sizes ($n=997$). Moreover, the study purpose is to identify key risk factors to prevent early dementia. AIC, deleting the variable starting from the full model, can fulfill this objective by precisely fitting, with the help of stepwise.

```
> stepAIC(null_m, scope=list(lower=null_m, upper=full_m), direction="both", k = 2)
```

Coefficients:	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-21.35437	3.27417	-6.522	6.93e-11 ***
Age	0.12453	0.02498	4.985	6.20e-07 ***
BMI	0.40217	0.10061	3.997	6.41e-05 ***
SmokingEx-smoker	-0.01622	0.32444	-0.050	0.96012
SmokingCurrent	1.45331	0.45626	3.185	0.00145 **

Null deviance: 408.4 on 996 degrees of freedom

Residual deviance: 355.0 on 992 degrees of freedom **AIC: 365**

The results show the model retained age, BMI, and current smoking as significant predictors of dementia ($p < 0.001$ and $p < 0.01$, respectively), while ex-smoker status was not significant ($p = 0.96$). Although the residual deviance was slightly higher, the AIC improved from 370.9 to 365.

Moreover, model validation needs to be conducted by likelihood ratio tests (LRT).

H_0 : The selected model fits the data as the full model

H_1 : The selected module couldn't be represented as the full model

```
> anova(full_m, aic_model, test='LRT')
```

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	988	352.9		
2	992	355.0	-4 -2.0999	0.7174

H_0 : The null model fits the data as the selected model

H_1 : The null module couldn't be represented as the selected model

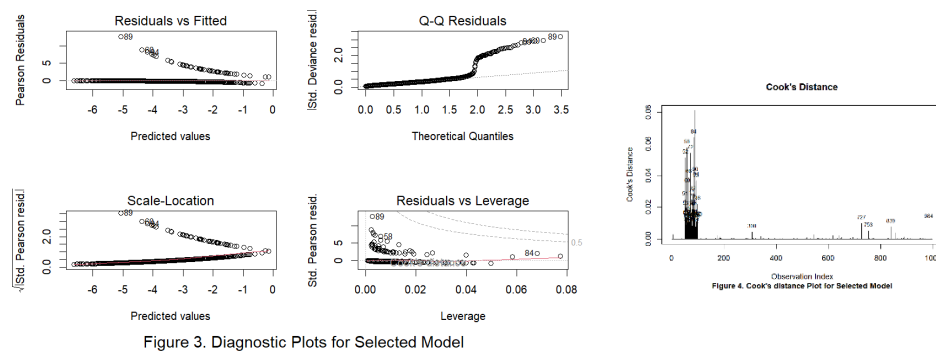
```
> anova(null_m, aic_model, test="LRT")
```

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	996	408.4		
2	992	355.0	4 53.402	7.02e-11 ***

Compared with the full model, the null hypothesis was not rejected ($p = 0.717$), indicating the excluded variable might not significantly improve the model's fitness. In contrast,

comparison with the null model yielded a highly significant result ($p < 0.001$), rejecting the null hypothesis, confirming that age, BMI, and smoking status provide substantial explanatory power for dementia risk. Therefore, the selection model is preferred because of balances fitness and simplifies.

3.3. Model Adequacy Checking



Model diagnosis was conducted to compare the model with the existing observations, willing to assess how well the model fits the data. Because of assumptions for neither normality nor constant variance on GLM, model checking for GLM is not as strict as it is for linear regression models. In Residuals versus Fitted, Q-Q Plot, and Residuals versus Leverage, even though the fitted lines match most of the observations, several extreme residuals suggest the model may underpredict dementia risk for certain individuals.

52 Influential observations are shown on Cook's Distance. When conducting sensitivity analysis by excluding observations exceeding the threshold of $4/n=0.004$, the resulting model yielded an AIC of 6. However, this approach led to the exclusion of all true dementia cases, rendering the statistic meaningless in this context. Therefore, we retain all observations in the final model.

3.4. Interpret Coefficients (Odds Ratios)

Table 3. Risk Factors for selected module

Variable	OR	95% CI	p-value
Age	1.13	(1.08 - 1.19)	0.0000006 ***
BMI	1.50	(1.23 - 1.83)	0.0000641 ***
SmokingEx-smoker	0.98	(0.51 - 1.85)	0.9601207
SmokingCurrent	4.28	(1.66 - 10.17)	0.0014463 **

Note:

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Tabel 3 illustrates the significant effect on the odds of dementia increasing by age and BMI, 13% and 50%, respectively. For a 10-year increase in age, the individual will have approximately 3.4 times ($1.13^{10} = 3.39$) odds of getting a dementia diagnosis. For every 5-unit increase in BMI, the odds of dementia increase 7.6-fold ($1.50^5 = 7.59$). For smoking status, ex-smokers showed no

significant difference in dementia risk compared with non-smokers (OR = 0.98). In contrast, current smokers had more than fourfold higher odds of dementia (OR = 4.28, $p < 0.01$), illustrating a substantial risk elevation.

To summarise, the AIC-selected logistic regression model is:

$$\begin{aligned} \text{logit}(P(\text{Dementia})) \\ = -21.35 + 0.125 \times \text{Age} + 0.402 \times \text{BMI} + 1.453 \times \text{Smoking}_{\text{Current}} \end{aligned}$$

The model demonstrates sufficient fit, highlighting age, BMI, and current smoking as significant predictors of dementia risk. These findings warrant further investigation in prospective studies to confirm their explanatory power and clinical relevance.

4) Investigate how reliable the dementia diagnosis is. (5 marks)

To evaluate the reliability of clinical dementia diagnosis, Table 4 shows the result of comparing doctor-diagnosed dementia against the true status of dementia using standard diagnostic metrics. The proportion for the comparison is demonstrated in Figure 5.

Table 4. Confusion Matrix: Clinical Diagnosis vs True Status

Clinical Diagnosis	True Dementia Status		
	No Dementia	Dementia	Total
Negative (No Dementia)	927	4	931
Positive (Dementia)	18	48	66
Sum	945	52	997

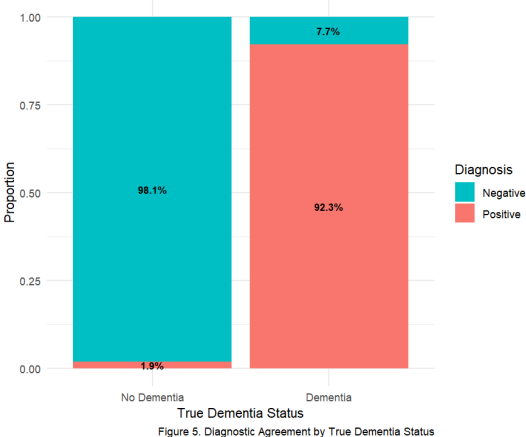


Figure 5. Diagnostic Agreement by True Dementia Status

Table 5. Diagnostic Test Performance Characteristics

Characteristics	Value	%	95% CI
Sensitivity	0.92308	92.3%	0.818 ~ 0.970
Specificity	0.98095	98.1%	0.970 ~ 0.988
PPV (Precision)	0.72727	72.7%	0.610 ~ 0.820
NPV	0.99570	99.6%	0.989 ~ 0.998
Accuracy	0.97793	97.8%	0.967 ~ 0.985

Table 5 calculated the important characteristics for the diagnostic test. These diagnosis methods have 92.3% sensitivity, meaning the test rarely misses actual dementia (only 4 false negatives out of 52 cases). Specificity, which is 98.1%, means very few healthy individuals are incorrectly diagnosed with dementia. When a doctor diagnoses dementia, there is a 72.7% chance the patient truly has the condition, which indicates Positive Predictive Value (PPV). Based on Bayes' theorem: $P[\text{Disease} \mid \text{Positive Test}]$, despite high sensitivity and specificity,

the PPV reflects the low prevalence of dementia (5.2%) in the sample. While a doctor rules out dementia, there is a 99.6% chance the patient is truly dementia-free, meaning Negative Predictive Value (NPV).

In brief, the diagnostic process demonstrated high reliability with 97.8% overall accuracy. However, due to the low prevalence and resulting risk of overdiagnosis, treatment strategies should be applied with caution.

5) Based on your findings, provide and justify your suggestions on what interventions could be possibly considered to prevent dementia. (8 marks)

Based on the previously selected model for early prevention of dementia, quitting smoking emerges as a primary priority. Current smokers have more than fourfold higher odds of dementia compared with non-smokers (OR = 4.28, $p < 0.01$).

Furthermore, reducing BMI is important. A one-unit decrease in BMI (kg/m^2) is associated with a 50% reduction in the odds of dementia (OR = 1.50, $p < 0.001$). To support weight control, individuals should aim to maintain BMI within the healthy range (18.5–24.9 kg/m^2) and monitor regularly to sustain fitness. Once the BMI exceeds this range, management strategies should include adjustments to daily diet and physical activity. Dietary recommendations involve reducing sugar and salt intake while avoiding processed foods and sweets. Moreover, at least exercising 150 minutes every week with a heart rate above 130 beats/min is crucial.

Third, each one-year increase in age was associated with a 13% higher odds of dementia (OR=1.13, $p<0.001$). Although age cannot be reversed, individuals can train their brains from an early age, such as memory games and cognitive exercises. At the same time, maintaining the **social network** is important because social engagement can stimulate and consolidate neuronal connections. Family members play a crucial role in detecting unusual changes in older adults. It might be useful to increase the screening frequency for the elderly, but the balance between cost and effectiveness needs to be evaluated.

To be brief, smoking cessation is the most cost-effective way to prevent early dementia. Maintaining BMI within the healthy range is also important, while cognitive training and social engagement may help mitigate age-related risk.

6) If you were told one of the variables identified as dementia risk factors in your analysis was not previously found to be associated with dementia, how might you use this information to adjust your analysis? (No formal analysis, just provide ideas and reasons. 5 marks)

6.1. Study design

The sample representativeness must be critically evaluated. If the Greater Manchester cohort could not represent health-conscious individuals or differed systematically from other

populations, the observed association may not generalize. Regional variations in risk factors should also be considered, as factors identified in other studies may not apply universally. Moreover, measurement quality requires scrutiny on self-reported variables like sleep and physical activity may introduce bias, and the absence of key confounders such as socioeconomic status, medication history, or comorbidities could distort results.

6.2. Statistical overview

During the analysis, several potential biases may affect the results. First, adjust the significance setting by corrections, such as Bonferroni, and interpretation should not rely solely on p-values but also consider 95% confidence intervals. Second, sensitivity analyses are needed to ensure the robustness of model selection, for example, by comparing stepwise, forward, and backward procedures under different cutoffs. Finally, subgroup analyses should be conducted to examine whether predictors remain significant within specific populations. Imbalances across subgroups may influence the observed associations with dementia risk, and thus require careful evaluation.

6.3. To adjust this study analysis

To adjust the analysis, a comprehensive literature review should first confirm whether the factor has been mentioned in narrative reviews or discussion sections. Biological plausibility should be assessed to support the observed association. Findings should be reported conservatively, clearly labeling novel factors as exploratory. Prospective studies are essential to establish temporal precedence and causality, while replication in independent datasets or experimental studies (e.g., animal models) would provide external validation.

To conclude, although identifying new risk factors is promising, restricted statistical analyses, supported biological mechanisms, and further consolidated trials are required. Thus, the current findings should be interpreted with caution and validated through future prospective studies.

7) Discuss at least four factors that might affect the interpretation of your findings and how these factors could be improved in future. (10 marks)

7.1. Sample Selection Bias and Representativeness

The dataset was collected only from patients in Greater Manchester, so population characteristics in other regions need to be considered. Although the dataset contains 977 observations, the proportion of dementia cases is low ($n = 52$, 5.2%). The sampling method described as random may still include bias, as individuals with higher health awareness could be overrepresented. This circumstance means the results may not be generalisable to other areas or countries. Furthermore, the precision of odds ratios may be reduced by the limited case number. Future improvements could include recruiting more participants across England, using stratified sampling to ensure representativeness, and adopting a case-control design to increase the number of dementia cases.

7.2. Measurement bias and quality of datasets

Self-reported risk factors may introduce bias. For example, individuals may not accurately recall their sleeping hours, leading to rough or unrealistic numbers. For daily exercise, the collected data are coarse and not supported by objective parameters such as heart rate. In addition, the criteria for distinguishing true dementia patients are unclear. Measurement bias and misclassification may affect statistical outcomes and reduce significance power. To reduce bias in datasets, sleep duration could be monitored by wearable devices, and physical activity should include more detailed information. Likewise, BMI and exercise frequency could be recorded more precisely, for example, by minutes of activity.

7.3. Study design on Cross-sectional

Because temporal order is unclear, the cross-sectional design cannot establish cause and effect between risk factors and dementia. This means the results cannot be interpreted as risk factors contributing to dementia, and the possibility of reverse causality must be considered. Dementia may cause cognitive impairment, which leads patients to change their lifestyle. Thus, physical activity may decline and BMI increase. Furthermore, dynamic changes are more informative than a single time point, and accumulated exposure cannot be captured in this design. Therefore, interpretation of odds ratios must be cautious, indicating that intervention effects cannot be inferred. This limitation could be improved if the study design were changed to a prospective study. Implement repeated measurements to capture cumulative exposure (e.g., pack-years of smoking, long-term BMI trends).

7.4. Confounding factors and unmeasured variables

Unmeasured confounders may distort the observed associations. For example, socioeconomic status, medication history, or comorbidities (depression, diabetes, cardiovascular disease) could influence dementia risk but were not included in the current dataset. Future studies should collect and adjust for these variables using multivariable models or stratified analyses to reduce bias and improve validity.